🧬 **Title: Introduction to Biology — Cells, Genetics, and Life**

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1. What is Biology?

Biology is the study of living organisms and life processes.
It focuses on how living things grow, function, reproduce, and interact with their environment.

Living organisms include:

- Animals
- Plants
- Bacteria
- Fungi

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2. The Cell — Basic Unit of Life

All living organisms are made of cells.

There are two main types of cells:

2.1 Prokaryotic Cells

- No nucleus
- Simple structure
- Example: bacteria

2.2 Eukaryotic Cells

- Have nucleus
- More complex
- Example: animal and plant cells

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3. Cell Organelles and Functions

Nucleus

Controls the activities of the cell and contains DNA.

Mitochondria

Produces energy through respiration.

Ribosomes

Responsible for protein synthesis.

Cell Membrane

Controls what enters and leaves the cell.

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4. DNA and Genetics

DNA carries genetic information in all living organisms.

Key concepts:

- DNA is made of nucleotides
- Genes are segments of DNA
- Genes control traits like eye color and height

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5. Protein Synthesis

Protein synthesis occurs in two steps:

Step 1: Transcription

DNA is converted into mRNA in the nucleus.

Step 2: Translation

mRNA is used to build proteins in ribosomes.

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6. Photosynthesis (Plants)

Photosynthesis is the process where plants make food using sunlight.

Equation:
$CO_2 + H_2O$ + light → glucose + oxygen

Occurs in chloroplasts.

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7. Cellular Respiration

Cells break down glucose to produce energy (ATP).

Occurs in mitochondria.

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8. Classification of Living Organisms

Living organisms are classified into:

- Kingdom Animalia
- Kingdom Plantae
- Kingdom Fungi
- Kingdom Protista
- Kingdom Monera

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9. Animals Overview

Animals are multicellular organisms that:

- cannot produce their own food
- depend on other organisms for energy
- have specialized organ systems

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10. Levels of Biological Organization

Cell → Tissue → Organ → Organ System → Organism